

Devdoots Assignment 07: Cyberpunk Logic Arcade & OOPs Quest

Team: Devdoots | **Team Leader:** Divyansh
Theme: Creative Logic Building, Dynamic Data Handling & Interactive OOPs
Submission Deadline: Wednesday, August 12, 2026, before 9:00 PM IST



STRICT ZERO-AI PROTOCOL & LIVE CODE DEFENSE

NO AI ALLOWED: ChatGPT, Gemini, Copilot ya kisi bhi AI / Vibe Coding tool ka use **100% STRICTLY PROHIBITED** hai.

MANDATORY LIVE VIVA: GitHub par Pull Request (PR) raise karne ke baad, har team member ko screen share karke apna code **Line-by-Line (ek-ek line ka logic)** explain karna hoga.

RULE: Agar koi member apni kisi bhi line ka logic explain nahi kar paaya, toh unka pura assignment **REJECTED** mana jayega.

1. The Badge & Reward System (Dopamine Unlocks)

Har task ko khud se solve karne aur code explain karne par team members ye official tags/badges unlock karenge:

Completed Level	Unlocked Title / Tag	Requirement
Task 1 Clear	 Cipher Master	Encryption & Decryption Algorithm bina kisi error ke banane par.
Task 2 Clear	 System Architect	Cart Billing Engine, Coupon Discounting & Tax Addition logic complete karne par.
Task 3 Clear	 Swarm Alchemist	Interactive Mini-DAIVY Cyber-Pet Simulator Class banane aur edge cases handle karne par.
All Tasks + Viva	 Devdoot Grand Cyber-Engineer	Bina AI ki madad ke saara code line-by-line explain karke PR merge karwane par.

2. Assignment Tasks (Creative & Logic Building)

Task 1: The Devdoot Secret Cipher Engine (String & ASCII Logic)

Goal: Team communication ke liye secret messages ko encrypt aur decrypt karne wala function banana.

- `encrypt_message(text, shift)`: Function banayein jo diye gaye text ke har letter ki ASCII value ko shift se aage badha de.
- `decrypt_message(encrypted_text, shift)`: Function banayein jo encrypted text ko wapas original message me convert kare.
- **Example Input:** text = "DEVDOOT", shift = 3
- **Expected Encrypted Output:** "GHYGRRW"
- **Creative Twist:** Spaces ya special characters (e.g. !, @) change nahi hone chahiye!
- **Unlocked Badge:**  Cipher Master

Task 2: Smart Cyber-Mart Billing Engine (Data Structures & Loops)

Goal: E-commerce cart aur billing system ka logic banana jo coupon codes aur GST tax calculate kare.

- Cart Data: cart = [{"item": "Sensor", "price": 1200, "qty": 2}, {"item": "Microcontroller", "price": 2500, "qty": 1}, {"item": "Motor", "price": 450, "qty": 4}]
- Function `generate_bill(cart, coupon_code)` banayein:
 - Subtotal count karein (price * qty).
 - Agar coupon_code == "DEVDOOT10" ho, toh 10% discount dein.
 - Discounted amount par 18% GST jodkar final amount calculate karein.
- Aakhir me ek clean receipt string format karke return karein.
- **Unlocked Badge:**  System Architect

Task 3: Mini-DAIVY Cyber-Pet / Bot Simulator (Interactive OOPs)

Goal: MiniDAIVY naam ki Class banana jo ek digital robot ki tarah behave kare.

- `__init__(self, bot_name)`: Bot ka naam set karein aur initial stats: energy = 100, battery = 100, mood = "HAPPY" set karein.
- Methods to Implement:
 - `do_task(task_name)`: Task karne par energy 20 kam ho aur battery 15 kam ho.
 - `charge_battery()`: Battery wapas 100% ho jaye.
 - `get_status()`: Bot ke current stats clean print kare.
- **Edge Case Logic:** Agar battery <= 10 ho jaye, toh bot koi task karne se mana kar de aur alert message de: *"System Low! Please charge Mini-DAIVY."*
- **Unlocked Badge:**  Swarm Alchemist

3. GitHub Submission Workflow & Single-Folder Rules

Step	Git Command	Description
1. Sync Main	git checkout main && git pull origin main	Latest main branch sync karein.
2. Create Branch	git checkout -b	Apni distinct feature branch

Step	Git Command	Description
	feature/yourname-assign7	banayein.
3. Create Folder & File	Folder: assignment_07/ File: yourname_assign7.py	Apne local repo me assignment_07 folder ke andar file banayein.
4. Single Folder Stage	git add assignment_07/	CRITICAL: git add . mat chalana, sirf assignment_07 folder add karein.
5. Commit & Push	git commit -m "[HW-7] Cyberpunk Arcade Attempt" git push origin feature/yourname-assign7	Commit karke branch remote par push karein.
6. Raise PR	GitHub Portal -> Create Pull Request	PR raise karke group me notify karein.

4. Post-Submission Viva & Code Defense Process

- PR raise hone ke baad, team meeting me har member screen share karega.
- Leader aur peers har line par questioning karenge: *"Yahan ASCII shift kaise ho raha hai?"*, *"Cart Loop me discount condition kaise apply hui?"*, *"Class me battery check method ke andar kaise ho raha hai?"*
- Un-aided, self-explained logic hone par hi PR merge ki jayegi aur Unlocked Badges award kiye jayenge.